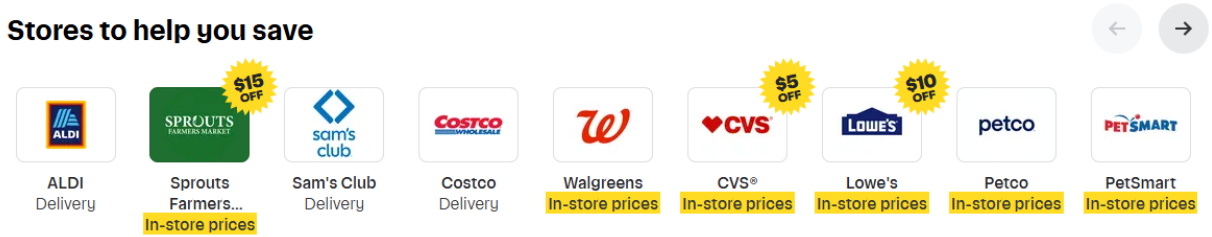
Mini project 3: Project: Online Supermarket Database System



Presentations deadline: April 21 and 23

No presentation, no grade.

Deliverables (due in 1 week)

Submit a `.zip` file containing:

1. `ER_Diagram.pdf` – A clear E-R diagram
2. `schema.sql` – MySQL script with table definitions
3. `data.sql` – Insert statements with test data
4. `queries.sql` – Required SQL queries
5. `php/` – Folder with PHP scripts for interfaces
6. `README.txt` – Instructions to run the project locally

% participation	ER Diagram	Tables Creation	Database design	PHP interfaces	Report	DB Population	Queries
Participant 1							
Participant 2							
Participant 3							
Participant 4							

No free riders, this time grade will be assigned based on active participation.

Overview

This project simulates the development of a relational database for an online supermarket. The system supports online product sales, tracks inventory and manages restock requests with suppliers. You will design the system using the E-R model, implement it in MySQL, populate it with test data, and create PHP interfaces for basic interactions.

Scope and Simplifications

- You will model a **single online store** (no geographic variation)
- Products are sold with static prices and quantities
- Focus on product sales, customer orders, inventory updates, and supplier requests

Requirements

1. E-R Diagram

- Design an E-R diagram with these core entities:

  i. `Customer`: who places orders
  ii. `Product`: available items with stock and price
  iii. `Order`: one per customer transaction
  iv. `OrderItem`: linking products with orders
  v. `Supplier`: restocks products
  vi. `RestockRequest`: request to supplier for more stock

- Show all primary keys and relationship cardinalities
- Use simple notation (you may draw this by hand or with a diagram tool)

2. Relational Schema and SQL Implementation

- Convert your E-R diagram into a relational schema
- Normalize your schema up to **3NF or BCNF**
- Implement tables in **MySQL**
- Include all relevant constraints:
  - Primary and foreign keys
  - `NOT NULL`, `CHECK` (if applicable)

3. Populate the Database

- Add **sample data** using SQL INSERTs:
  - At least 5 customers
  - 10 products
  - 3 suppliers
  - 10 orders with multiple items
- Data may be realistic or randomly generated

- Feel free to add more entities or relationships, remove them, or modify the ones already listed to better fit your design your design idea.

4. Sample SQL Queries

Write and test the following queries:

- Find the 5 top-selling products
- Find products below the restock threshold
- Find which customers placed the most orders
- List products ordered in a given date range

5. PHP Interfaces

Create a **basic PHP interface** (no CSS or frameworks needed) that allows:

- Basic interface with product categories (sodas, bread, etc), customer click on category place an order.
- Customers to place an order (interface)
- Customer check-out (customer select the number of products and pay for them)
- The system to update inventory and update the supermarket banking account balance.
- Automatic restock requests for low-stock products
- Suppliers to confirm delivery and restock inventory

You can use simple HTML forms and PHP to connect to the MySQL database.

Grading Criteria (100 points)

Component	Points
E-R Diagram	20
SQL Schema & Constraints	20
Test Data Creation	10
Sample SQL Queries	10
PHP Interfaces (Functional, Simple)	30
Code Organization & Documentation	10

BONUS (Extra 10 points)

Shoping car nice desing. The customer can pick a variety of product and pay at the end. Below an example.

Shopping Cart

Deselect all items

Price

Bronson Vitamin D3 10,000iu (250mcg) 1 Year Supply for Healthy Muscle Function and Immune Support, Non-GMO, 360 Tablets

In Stock

✓prime Overnight

FREE delivery **Overnight 7 AM - 11 AM** on $25 of qualifying items

This is a gift Learn more

Flavor Name: Unflavored

Size: 360 Count (Pack of 1)

1 | Delete | Save for later | Compare with similar items | Share

Limited time deal

-10% $8.97

($0.02 / Count)

Typical price: ~~$9.97~~

Save 5% now with Subscribe & Save ›

Save 5% on 2 select item(s)

Shop items

NORTIV 8 Men's Walking Shoes Running Tennis Cushioning ActiveBreeze Non Slip Athletic Gym Workout Comfortable Fashion Sneakers,Size 8,Blue,SNWS246M

In Stock

✓prime Two-Day

FREE delivery **Sat, Apr 12**

FREE Returns

This is a gift Learn more

Size: 8

Color: Blue

1 | Delete | Save for later | Compare with similar items | Share

$52.24

Subtotal (2 items): **$61.21**

$25

Get FREE Overnight delivery with $16.03 more of eligible items. Find eligible items ›

Subtotal (2 items): **$61.21**

This order contains a gift

Proceed to checkout

Complete your basket with these items

Bronson Vitamin B 10...

2,118

$9.97 ($0.17/Count)

✓prime **Overnight by 11:00 AM**

Add to cart

Bronson Ashwagandh...

5,688

$9.99 ($0.08/Count)

✓prime **Overnight by 11:00 AM**

Add to cart